

Instructions for running tutorials in HelioCloud



These instructions are here:

https://github.com/heliocloud-data/science-tutorials/tree/gem/gem2026/Heliocloud_Setup.pdf

Create a HelioCloud account

Go to: <https://daskhub-demo.heliocloud.org/>

Click on the small “Sign Up” link.

A screenshot of the HelioCloud sign-in page. It features a title "Sign in with your username and password", input fields for "Username" and "Password", a "Forgot your password?" link, a blue "Sign in" button, and a link "Need an account? Sign up" at the bottom. The "Sign up" link is circled in red.

This then is the sign-up page.

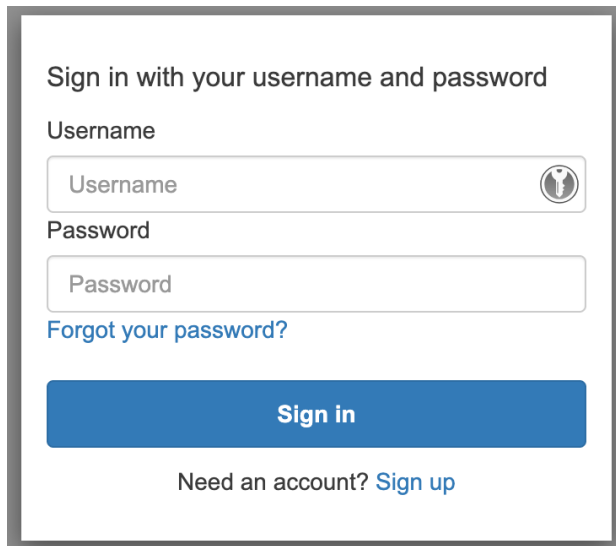
A screenshot of the HelioCloud sign-up page. It features a title "Sign up with a new account", input fields for "Username", "Email", and "Password", a blue "Sign up" button, and a link "Already have an account? Sign in" at the bottom.

Registration is simple. Just enter your email and choose a username and password.

When you sign up, it also logs you in that first time. For future sessions you will need to log in.

Logging in to HelioCloud (for usage after the first time)

Go to: <https://daskhub-demo.heliocloud.org/> and enter your username and password.

A login form for HelioCloud. It has a title "Sign in with your username and password". Below it are two input fields: "Username" and "Password". The "Username" field has a placeholder text "Username" and a small icon of a person. The "Password" field has a placeholder text "Password". Below the "Password" field is a link "Forgot your password?". At the bottom is a blue button labeled "Sign in". Below the button is a link "Need an account? Sign up".

Sign in with your username and password

Username

Username

Password

Password

[Forgot your password?](#)

Sign in

Need an account? [Sign up](#)

Choose a virtual machine instance type and the default is fine: "Demo Server (CPU)". If you want to try GPU computing, choose the other option, "Demo Server (GPU)". Click the "Start" button.

It can take several minutes for your environment to get created. You should see a progress bar and some scrolling info messages. Some may mention warnings, which are not a problem. If this step fails, try logging in again. (On rare occasions, Amazon does not have enough capacity to start another instance of the type requested.)

Server Options

☒ **Demo Server (CPU)**

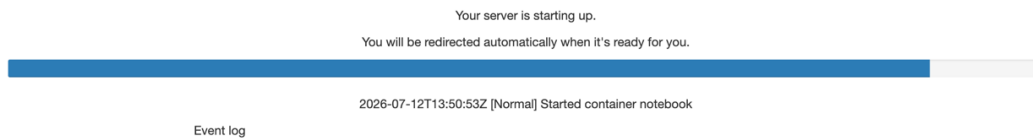
Small notebook server for demos. 8GB RAM/2 CPU. ~\$0.10/hr

☐ **Demo Server (GPU)**

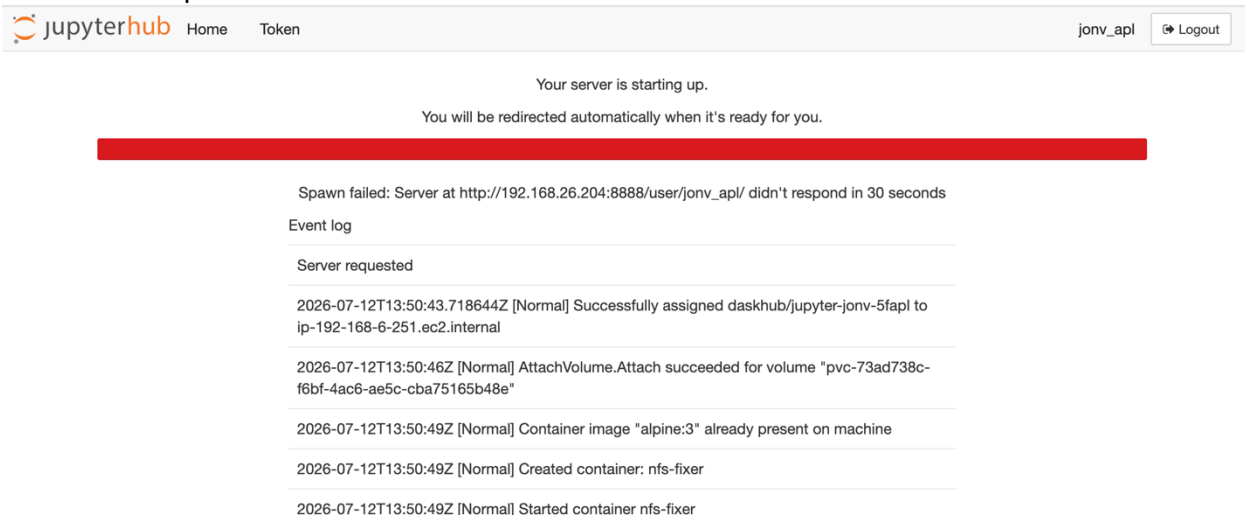
Small GPU server for ML demos. NVidia T4 GPU, 16GB RAM/4 CPU. ~\$0.40/hr

Start

Normal login shows progress like this:

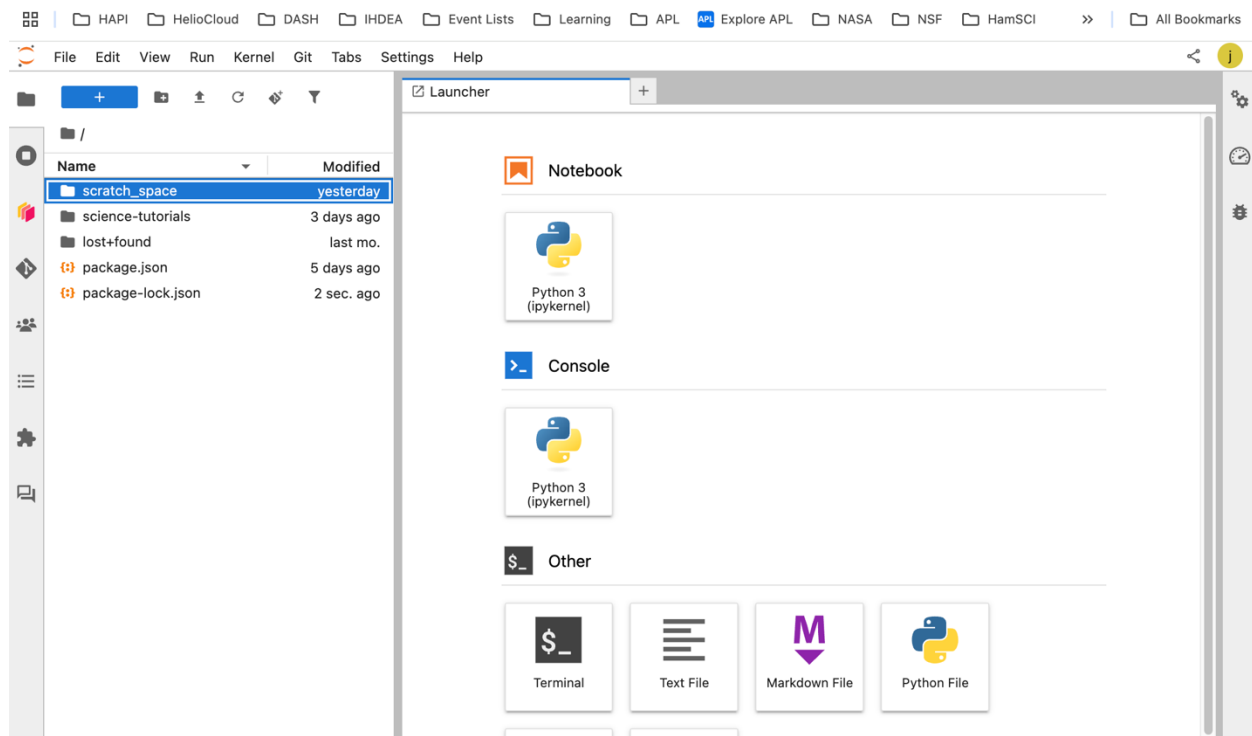


A failed startup looks like this:



Click the Jupyter Hub icon on the top left, and then try again.

Once your virtual server starts, you will see an environment like this:



This is showing you cloud-based workspace environment. It stores content with the following directory structure (the list of files and directories on the left):

science-tutorials – this is a copy (checked out from Git) of a set of pre-made routines that illustrate how to use HelioCloud. It contains Python notebooks and some regular Python scripts, all focused on tutorial activities.

scratch_space – this is your own working area

Ignore these directories (they are part of the Unix and container infrastructure):

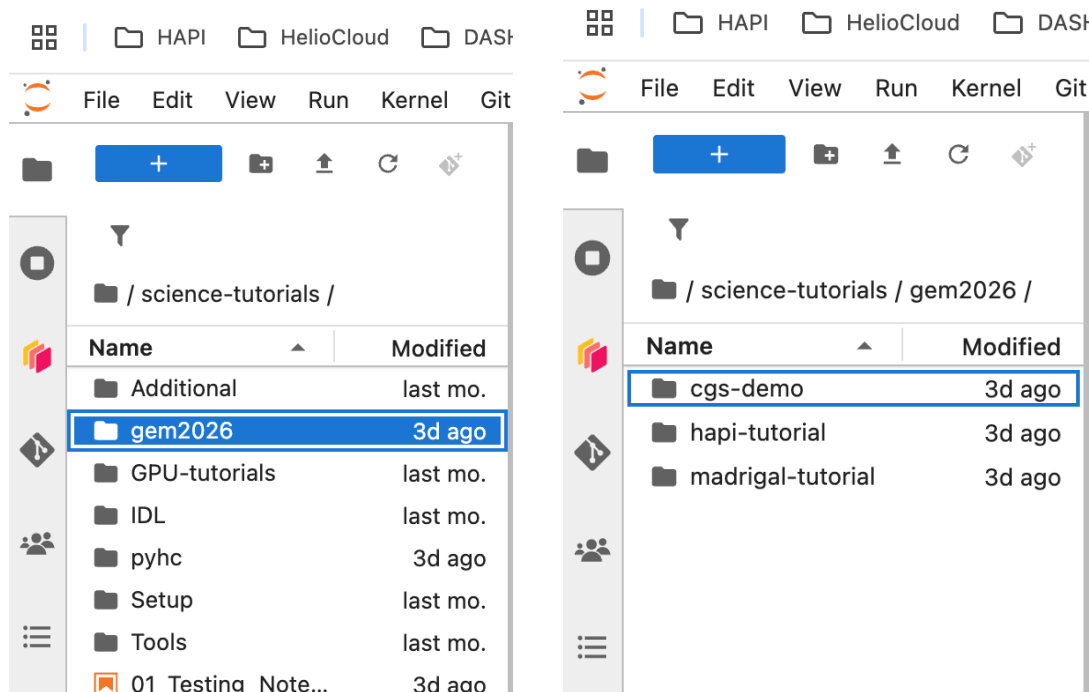
lost+found – ignore this (Unix infrastructure)

package.json

package-lock.json

You can open a terminal window by clicking the Terminal” icon or start a Python Notebook using the Python 3 Notebook icon (first one in the list).

Inside the **science-tutorials** directory, there is another directory called **gem2026**. This has all the HelioCloud demos for the GEM meeting.



Open the directory with the tool of interest, and start exploring the notebooks.

Points to emphasize:

The entire **science-tutorials** directory is your own copy. You can change that code (and running a notebook will change it, since it adds or modifies the output). You will not be able to check those changes back into Git.

The **scratch_space** areas is shared by all users, so you can use that for collaboration with other users.